

Healthcare Facilities Performance Analysis Dashboard

Executive Summary

The Healthcare Facilities Performance Analysis Dashboard was developed to evaluate healthcare infrastructure, workforce capacity, service availability, and facility quality across Pakistan. The project analyzes healthcare facilities located in major cities and provinces, providing insights into healthcare resources, staffing levels, emergency service coverage, and patient satisfaction ratings.

The dashboard transforms raw healthcare data into actionable business intelligence, enabling stakeholders to identify resource gaps, assess healthcare readiness, and support data-driven decision-making.

Business Problem

Healthcare administrators and policymakers require accurate information about healthcare facilities to ensure efficient resource allocation and quality service delivery. Understanding the distribution of healthcare infrastructure, workforce availability, emergency services, and facility ratings is essential for improving healthcare outcomes.

This project aims to provide a comprehensive overview of healthcare performance and identify opportunities for operational and strategic improvements.

Dataset Overview

The dataset contains healthcare facility information from multiple cities and provinces across Pakistan.

Key Data Fields

- Facility Name
- City
- Province
- Facility Type
- Ownership Type (Public / Private)
- Total Beds
- Total Doctors
- Total Nurses
- Total Staff
- Emergency Service Availability
- Facility Rating

The dataset includes hospitals, clinics, medical centers, maternity homes, and diagnostic laboratories located throughout the country.

Business Questions

Infrastructure Analysis

1. How many healthcare facilities are operating?
2. Which province has the highest number of facilities?
3. Which cities have the strongest healthcare infrastructure?
4. What is the total bed capacity available?

Workforce Analysis

1. How many doctors support healthcare services?
2. How many nurses are available across facilities?
3. What is the total healthcare workforce capacity?
4. Which provinces have the highest concentration of healthcare professionals?

Facility Performance Analysis

1. Which facility types are most common?
2. What percentage of facilities are public versus private?
3. Which facility type receives the highest ratings?
4. Which facilities demonstrate the best performance?

Emergency Service Analysis

1. What percentage of facilities provide emergency services?
2. Which provinces have the strongest emergency coverage?
3. Are emergency services evenly distributed across regions?

Quality Analysis

1. What is the average facility rating?
 2. Which provinces have the highest-rated facilities?
 3. Which facility types provide the highest quality services?
 4. What factors influence facility ratings?
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Analysis Performed

Facility Distribution Analysis

Healthcare facilities were analyzed by city, province, and facility type to understand infrastructure distribution across the country.

Workforce Capacity Analysis

The availability of doctors, nurses, and support staff was evaluated to measure healthcare workforce strength.

Bed Capacity Analysis

Hospital bed availability was assessed to understand healthcare capacity and readiness.

Emergency Service Assessment

Emergency service availability was analyzed to identify regions with stronger emergency healthcare coverage.

Facility Quality Evaluation

Facility ratings were examined to assess healthcare quality and identify top-performing facilities.

Key Findings

Healthcare Infrastructure

The healthcare system consists of a diverse network of hospitals, clinics, diagnostic labs, maternity homes, and medical centers distributed across multiple provinces.

Workforce Capacity

Healthcare services are supported by significant numbers of doctors, nurses, and support staff, demonstrating the importance of workforce planning in healthcare operations.

Emergency Readiness

Emergency services are available in many facilities; however, coverage levels vary across regions, highlighting potential opportunities for service expansion.

Quality Performance

Facility ratings indicate variations in service quality among healthcare providers. High-performing facilities can serve as benchmarks for quality improvement initiatives.

Regional Differences

Healthcare resources and facility performance differ between provinces and cities, suggesting the need for targeted investment strategies.

Business Recommendations

Resource Allocation

- Increase investment in underserved regions.
- Expand healthcare capacity where bed availability is limited.

Workforce Planning

- Improve recruitment and retention of healthcare professionals.
- Balance workforce distribution across provinces.

Emergency Services

- Expand emergency service coverage in low-coverage areas.
- Improve emergency preparedness and response capabilities.

Quality Improvement

- Monitor low-rated facilities and implement performance improvement plans.
- Adopt best practices from high-performing facilities.

Strategic Planning

- Use healthcare performance metrics to support long-term infrastructure development.
- Continuously monitor facility performance through KPI dashboards.

Skills Demonstrated

- Data Cleaning
- Data Transformation
- Data Modeling
- Power BI Dashboard Development
- KPI Reporting
- Data Visualization
- Business Intelligence
- Healthcare Analytics
- Trend Analysis
- Data Storytelling

Conclusion

The Healthcare Facilities Performance Analysis Dashboard provides a comprehensive view of healthcare infrastructure, workforce resources, emergency service coverage, and facility quality across Pakistan.

Through interactive visualizations and KPI reporting, the project enables healthcare stakeholders to make informed decisions that improve operational efficiency, service quality, and healthcare accessibility.